MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 12AV7

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, medium Mu

© Outline --- 6-2 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	2a	2g	2k	h	h	1a	1g	1k	hct

ABSOLUTE-MAXIMUM RATINGS:

Parameter	Ef	Eb	Ec	Ehk	Rk/k	Pp/p	© Alt
Unit:	V	Vdc	Vdc	v	Ohms	W	ft
Maximum	6.9 (prl) or 13.9 (ser)	330	---	100	---	3	(see note 1)
Minimum:	5.7 (prl) or 11.3 (ser)	---	---	---	---	---	---
TEST CONDITIONS:	12.6 (ser)	150	0	---	56	---	---

GENERAL:

Qualification - Not required

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Ec = -1 Vdc; Rk = 27 ohms (see notes 3 and 4)	0.65	II	Ic	0	-3.0	µAdc
1306	Transconductance (1)	See note 5	0.65	II	Sm	6,800	10,000	µmhos
1231	Emission	Eb = Ec = 10 Vdc; Rk = 0 (see notes 2, 3, and 6)	0.65	II	Is	50	---	mAdc
1256	Electrode current (1) (anode)	See note 5	0.65	II	Ib	13	23	mAdc
1256	Electrode current (2) (anode)	Ec = -15 Vdc; Rk = 0 (see note 6)	0.65	II	Ib	---	50	µAdc
(C) 1201	Short and discontinuity detection	Short and discontinuity	0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1301	Heater current		6.5	S4	If	207	243	mA
(C) 1336	Heater-cathode leakage	See note 4	6.5	S4	Ihk	---	30	µAdc
1306	Transconductance (2)	Ef = 11.0 V (see note 5)	6.5	S4	Sm	6,000	---	µmhos
1306	Transconductance (3)	Eb = 100 Vdc; Rk = 120 ohms (see note 5)	6.5	S4	Sm	4,800	7,300	µmhos
(C) 1316	Amplification factor	See note 5	6.5	S4	Mu	37	52	---
1331	Direct-interelectrode capacitance	Shield No. 315	6.5	S4	Cgp	1.5	2.3	pF
					Cin	2.3	3.5	pF
					Cout 1	0.9	1.7	pF
					Cout 2	1.2	2.0	pF
					Cpp	---	0.4	pF
1031	Low-frequency vibration	Rk = 0; Ec = -2 Vdc; Rp = 2,000 ohms (see note 2)	6.5	S4	Ep	---	150	mVac
(C) 1105	Permanence of marking		---	---	---	---	---	---
(C) 1211	Insulation of electrodes		4.0	S3	---	---	---	---
	Quality conformance inspection, part 3							
1501	Intermittent life	Group A; Eb = 100 V; Rk = 29 ohms (see note 4)	---	---	---	---	---	---
----	Intermittent life-test end points (500 hours)	Transconductance (1)	---	---	Sm	5,800	---	µmhos
		Electrode current (grid)	---	---	Ic	0	-4.5	µAdc

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. Tests may be performed with filament series connected for 12.6 V or parallel connected for 6.3 V.
3. This test to be performed at the conclusion of the holding period.
4. Test with sections connected in parallel.
5. Test each unit separately but with test voltages applied to both units.
6. Test each section separately with opposite section grounded.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI
Navy -
Air Force - 11, 85
DSA - ES

User activities:
Army - WC, AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-2)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the s ification in procurement of products for ultimate use by the Department of Defense. This sheet is provided fo taining information on the use of this specification which will insure that suitable products can be procured w minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fol lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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